Caspase-like activity is associated with bacterial infection of the urine in urinary tract diseases.

PMID: 38280678 | Indexed in PubMed

Urinary tract infections (UTIs) are a serious public health problem. They can be caused by a number of pathogens, but the most common are Escherichia coli, Klebsiella pneumoniae, Proteus mirabilis, Enterococcus faecalis and Staphylococcus saprophyticus. Bacterial infection is diagnosed by examining a urine sample. The presence of bacteria or white blood cells is determined under a microscope or a urine culture is performed. In this study, we used a panel of chromogenic substrates for the qualitative determination of specific enzyme activity in the urine of patients with confirmed bacterial infection and/or urinary tract disease. Healthy patients were used as a control group. It turned out that in the case of Escherichia coli infection, we observed the activity of the caspase subunit of the human 20S proteasome. We did not observe similar correlations for infections with other types of bacteria.

Bacterial resistance to antimicrobials in urinary isolates.

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Escherichia coli accounted for about 80% of organisms in uncomplicated urinary tract infections (UTIs), followed by Staphylococcus spp. especially Staphylococcus saprophyticus, and Proteus mirabilis. Against E. coli isolates from patients with uncomplicated UTI, faropenem was the most effective. Up to 1999, fluoroquinolone-resistant isolates were not observed in patients with uncomplicated UTI, but in 2001 fluoroquinolone-resistant E. coli isolates emerged and accounted for about 8%. Various types of organisms were isolated in patients with complicated UTI. Enterococcus faecalis, E. coli, and Pseudomonas aeruginosa were the three most frequent organisms isolated. These three organisms accounted for 44.6%. Amongst oral agents, faropenem showed the lowest rate of resistance against E. coli followed by cephems. The rates of highly fluoroquinolone-resistant and cefpodoxime-resistant E. coli isolates increased rapidly from 1998 to 2001. Fluoroquinolone-resistant P. aeruginosa isolates accounted for about 40% in 2001. Against this species, amikacin was the most effective antimicrobials among all agents tested. About 17% of Pseudomonas were resistant to carbapenem. Eight milligram per litre of ampicillin inhibited all E. faecalis isolates; about 60% of Enterococcus faecium were resistant to ampicillin. The rates of levofloxacin-resistant isolates of E. faecalis and E. faecium were 38 and 97% respectively. UTIs caused by vancomycin resistant enterococci (VRE) are rare in Japan.

Hidden pathogens uncovered: metagenomic analysis of urinary tract infections.

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Urinary tract infections (UTIs) are the most common kidney and urologic diseases in industrial nations and are usually caused through faecal contamination of the urinary tract. In this study, we have examined 1449 urine specimens both by culture and by PCR. The majority of UTIs examined were caused by Escherichia coli (35.15%), followed by miscellaneous bacteria (23.03%), and by Enterococcus faecalis (19.39%). A large fraction of fastidious and anaerobic bacteria (22.43%) was not detected under culture conditions but only by using PCR. This group of bacteria evade the standard culture conditions used in routine diagnostic laboratories examining urine specimens. The molecular approach used broad-range

16S rDNA PCR, denaturing high-performance liquid chromatography analysis, sequencing, and bioinformatic analysis to uncover these 'hidden' pathogens and is recommended in particular when examining leukocyte esterase-positive and culture-negative urinary tract specimens.

[Research on mechanisms of drug resistance and virulence expression in pathogenic Escherichia coli].

PMID: 37880099 | Indexed in PubMed

Urinary tract infections (UTIs) are one of the most common infections. Uropathogenic Escherichia coli (UPEC) is the most common causative organism. Once UPEC enters the urinary tract, it infects the bladder and then ascends the urinary tract to the kidneys, where it causes pyelonephritis, a more severe form of the disease. While various virulence factors, including adhesions and cytotoxic factors to bladder epithelial cells, have been identified and their functions have been analyzed, the question remains, "How can UPEC, which is harmless in the intestinal tract, be induced to become pathogenic in the urinary tract?" and "How does UPEC ascend the urinary tract and infect the kidneys?" On the other hand, UPEC invades host cells and forms biofilm-like microcolonies that are resistant to various antimicrobial agents. We are working to solve this problem by identifying the factors responsible for the virulence of UPEC and the establishment of infection of the kidney, as well as the factors involved in microcolony formation and elucidating their functions. Here I outline the virulence expression of UPEC from bladder to kidney infection and the mechanism of UTI refractoriness, focusing on our studies.

Expression of flagella is coincident with uropathogenic Escherichia coli ascension to the upper urinary tract.

PMID: 17925449 | Indexed in PubMed

Uropathogenic Escherichia coli (UPEC) cause most uncomplicated urinary tract infections (UTIs) in humans. Because UTIs are considered to occur in an ascending manner, flagellum-mediated motility has been suggested to contribute to virulence by enabling UPEC to disseminate to the upper urinary tract. Previous studies from our laboratory and others have demonstrated a modest yet important role for flagella during ascending UTI. To better understand the role of flagella in vivo, we used biophotonic imaging to monitor UPEC infection and temporospatial flagellin gene expression during ascending UTI. Using em7-lux (constitutive) and fliC-lux transcriptional fusions, we show that flagellin expression by UPEC coincides with ascension of the ureters and colonization of the kidney. The patterns of fliC luminescence observed in vitro and in vivo were also validated by comparative quantitative PCR. Because fliC expression appeared coincident during ascension, we reassessed the contribution of fliC to ascending UTI using a low-dose intraurethral model of ascending UTI. Although wild-type UPEC were able to establish infection in the bladder and kidneys by 6 hours postinoculation, fliC mutant bacteria were able to colonize the bladder but were significantly attenuated in the kidneys at this early time point. By 48 hours postinoculation, the fliC mutant bacteria were attenuated in the bladder and kidneys and were not detectable in the spleen. These data provide compelling evidence that wild-type UPEC express flagellin and presumably utilize flagellum-mediated motility during UTI to ascend to the upper urinary tract and disseminate within the host.

Kinetics of uropathogenic Escherichia coli metapopulation movement during urinary tract infection.

PMID: 22318320 | Indexed in PubMed

The urinary tract is one of the most frequent sites of bacterial infection in humans. Uropathogenic Escherichia coli (UPEC) strains are the leading cause of urinary tract infections (UTIs) and are responsible for greater than 80% of uncomplicated cases in adults. Infection of the urinary tract occurs in an ascending manner, with colonization of the bladder leading to possible kidney infection and bacteremia. The goal of this study was to examine the population dynamics of UPEC in vivo using a murine model of ascending UTI. To track individual UPEC lineages within a host, we constructed 10 isogenic clones of UPEC strain CFT073 by inserting unique signature tag sequences between the pstS and glmS genes at the attTn7 chromosomal site. Mice were transurethrally inoculated with a mixture containing equal numbers of unique clones. After 4 and 48 h, the tags present in the bladders, kidneys, and spleens of infected mice were enumerated using tag-specific primers and quantitative real-time PCR. The results indicated that kidney infection and bacteremia associated with UTI are most likely the result of multiple rounds of ascension and dissemination from motile UPEC subpopulations, with a distinct bottleneck existing between the kidney and bloodstream. The abundance of tagged lineages became more variable as infection progressed, especially after bacterial ascension to the upper urinary tract. Analysis of the population kinetics of UPEC during UTI revealed metapopulation dynamics, with lineages that constantly increased and decreased in abundance as they migrated from one organ to another. Urinary tract infections are some of the most common infections affecting humans, and Escherichia coli is the primary cause in most uncomplicated cases. These infections occur in an ascending manner, with bacteria traveling from the bladder to the kidneys and potentially the bloodstream. Little is known about the spatiotemporal population dynamics of uropathogenic E. coli within a host. Here we describe a novel approach for tracking lineages of isogenic tagged E. coli strains within a murine host by the use of quantitative real-time PCR. Understanding the in vivo population dynamics and the factors that shape the bacterial population may prove to be of significant value in the development of novel vaccines and drug therapies.

Pathogenesis of urinary tract infections: a review.

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Pathogenesis of urinary tract infections (UTIs) is not well-understood. In this paper, we review the current understanding of UTIs, particularly in relationship to individuals using intermittent catheterization. Relationships exist between the human host, infectious agent and the environment. In the human host, the urethra connects the bladder to potential infectious agents on the perineum. A high-pressure zone exists within the urethra at a point where the urethra passes through the urogenital diaphragm. This zone creates a natural barrier to ascent of organisms colonized in the distal urethra and the bladder itself has natural defences against invading organisms. The interaction of host defences with bacteria (infectious agent) determines whether or not the bacteria persist. A small number of bacteria and some types of bacteria are controlled more effectively by natural bladder defence mechanisms and frequent bladder emptying than a large number of bacteria. Escherichia coli, coliforms and enterococci are considered common bacterial causes of UTIs and are found in high numbers on the perineum. Intermittent catheterization is an effective way of bladder emptying but as an invasive procedure it remains a risk factor in the development of UTI.

Urinary Tract Infection among Renal Transplant Recipients in Yemen.

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Urinary tract infection (UTI) is the most common complication following kidney transplantation (KT), which could result in losing the graft. This study aims to identify the prevalence of bacterial UTI among KT recipients in Yemen and to determine the predisposing factors associated with post renal transplantation UTI. A cross sectional study included of 150 patients, who underwent KT was conducted between June 2010 and January 2011. A Morning mid-stream urine specimen was collected for culture and antibiotic susceptibility test from each recipient. Bacterial UTI was found in 50 patients (33.3%). The prevalence among females 40.3% was higher than males 29%. The UTI was higher in the age group between 41-50 years with a percentage of 28% and this result was statistically significant. Predisposing factors as diabetes mellitus, vesicoureteral reflux, neurogenic bladder and polycystic kidney showed significant association. High relative risks were found for polycystic kidney = 13.5 and neurogenic bladder = 13.5. The most prevalent bacteria to cause UTI was Escherichia coli represent 44%, followed by Staphylococcus saprophyticus 34%. Amikacin was the most effective antibiotic against gram-negative isolates while Ciprofloxacin was the most effective antibiotic against Staphylococcus saprophyticus. In conclusion, there is high prevalence of bacterial UTI among KT recipients in Yemen. Diabetes mellitus, vesicoureteral reflux, neurogenic bladder, polycystic kidney and calculi were the main predisposing factors.

Case Report: Escherichia fergusonnii - Pathogen in Urinary Tract Infection.

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Urinary tract infections are one of the leading cause of morbidity in admitted patients. Most commonly caused by Escherichia coli, but there are some variants which are commonly reported in urinary tract infection. This study was about to speciate such isolate like E.fergusonnii and find out its antibiogram.

FACTORS ASSOCIATED WITH EXTENDED SPECTRUM β-LACTAMASE PRODUCING ESCHERICHIA COLI IN COMMUNITY-ACQUIRED URINARY TRACT INFECTION AT HOSPITAL EMERGENCY DEPARTMENT, BANGKOK, THAILAND.

PMID: 27244961 | Indexed in PubMed

Urinary tract infection or UTI is most commonly caused by Escherichia coli. This study investigated the prevalence of and risk factors for extended spectrum β-lactamase-producing (ESBL) E. coli in community-acquired UTI presenting at the Emergency Department, Faculty of Medicine Ramathibodi Hospital, Mahidol University, Bangkok, Thailand. A retrospective review was conducted over a one-year period (2014) of case histories of patients over 15 years of age diagnosed with (n = 159) and without culture-positive (n = 249) ESBL E. coli. Backward stepwise multivariate logistic regression analysis revealed four independent risk factors for UTI caused by ESBL E. coli, namely, urinary catheter use, previous UTI in which ESBL E. coli was present, and previous use of antibiotics cephalosporin and penicillin. This information should be useful in devising future public health prevention and control programs for ESBL E. coli-associated community-acquired UTI.